

Question	Answer	Marks
8(a)	photoelectric effect	B1
8(b)(i)	$E = hf$	C1
	work function = $6.63 \times 10^{-34} \times 8.8 \times 10^{14}$ $= 5.8 \times 10^{-19} \text{ J}$	A1
8(b)(ii)	$hf = \phi + \frac{1}{2}mv_{\text{MAX}}^2$	C1
	$6.63 \times 10^{-34} \times 11 \times 10^{14} = (5.8 \times 10^{-19}) + (\frac{1}{2} \times 9.11 \times 10^{-31} \times v_{\text{MAX}}^2)$	C1
	$v_{\text{MAX}} = 5.7 \times 10^5 \text{ m s}^{-1}$	A1
8(c)	E_{MAX} shown as zero from $f = 8.0$ to 8.8 and non-zero from $f = 8.8$ to 11	B1
	all non-zero E_{MAX} shown as a single straight line with a positive gradient	B1
	line passing through (11, 1.45)	B1

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